# JoVALE: Detecting Human Actions in Video Using Audiovisual and Language Contexts (Supplementary Material)

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## Implementation Details

In this section, we provide the implementation details, including the hyperparameters, training and inference procedures, and the loss function employed in our JoVALE model.

### Hyperparameters

The Actor-centric Multi-modal Fusion Network (AMFN) consists of $L = 6$ layers. For actor proposals features, we extract the top $K$ proposals, where $K$ is set to 15. The structural details of the Multi-modal Feature Encoding (MFE) and Multi-modal Feature Aggregation (MFA) modules, which constitute the AMFN, are presented in Table 1 and Table 2, respectively. Table 1 outlines the hyperparameters of the MFE, while Table 2 provides details of the bottleneck MLP employed in the MFA. For temporal alignment and de-alignment in MFA, we set $T_c$ equal to $T_v$ and follow the approach (Cooper 2019), applying temporal average pooling when $T_{mod} \geq T_c$ and temporal repetition when $T_{mod} < T_c$.

| Hyperparameters | MFE |
|---|---|
| Embedding dimension | 256 |
| FFN hidden dimension | 1024 |
| Attention heads | 8 |
| Activation function | GELU |

Table 1: Hyperparameters of the MFE.

| Hyperparameters | MFA |
|---|---|
| Input dimension | 768 |
| FFN hidden dimension | 48 |
| Output dimension | 768 |
| Activation function | RELU |

Table 2: Hyperparameters of bottleneck MLP in the MFA.

The audio waveform samples, spanning $t$ seconds, are converted into spectrograms of size $100t \times 128$. These spectrograms are subsequently passed through a convolutional layer with a kernel size of $P = 16$ and a stride of $S = 10$, yielding audio embeddings with a temporal length of $T_a = 20$. For extracting scene-descriptive features, the number of input frames $T_s$ is set to 4. The temporal length of the visual features $T_v$ varies depending on the backbone architectures. $T_v$ is 4 when using SlowFast (Feichtenhofer et al. 2019) and 8 when using ViT (Dosovitskiy et al. 2020). The hyperparameter $D$, representing the channel width for all embeddings, is set to 256.

### Training and Inference

Table 3 provides details of the training settings and input resolutions for JoVALE on each dataset. The input resolution refers to the length of the shorter side of an input image, resized for visual feature extraction. For all datasets (Gu et al. 2018; Soomro, Zamir, and Shah 2012; Jhuang et al. 2013), images are resized to $384 \times 384$ for scene-descriptive feature extraction in both the training and inference phases. For data augmentation, we applied random horizontal flipping to RGB frames. For audio, we utilized SpecAugment (Park et al. 2019) with time and frequency masking, following the approach in AST (Gong, Chung, and Glass 2021).

### Loss Function

Our proposed model predicts action scores for $K$ actors detected by a pre-trained person detector (Ren et al. 2015), where the $K$ is set to be larger than the typical number of actors in a video clip. The predicted set $\hat{\mathbf{y}} = \{\hat{y}_i\}_{i=1}^{K} = \{(b_i, h_i, c_i)\}_{i=1}^{K}$, includes the predicted bounding box $b_i$, human score $h_i$, and action score $c_i$. Given that there are $K^{gt}$ ground truth labels $\{\tilde{y}_i\}_{i=1}^{K^{gt}} = \{(\tilde{b}_i, \tilde{h}_i, \tilde{c}_i)\}_{i=1}^{K^{gt}}$ in the keyframe, we construct the ground truth set $\mathbf{y^{gt}} = \{y_i^{gt}\}_{i=1}^{K} = \{(b_i^{gt}, h_i^{gt}, c_i^{gt})\}$, where

$$y_i^{gt} = \begin{cases} \tilde{y}_i, & \text{if } 1 \leq i \leq K^{gt} \\ (\varnothing, \varnothing, \varnothing), & \text{otherwise.} \end{cases} \quad (1)$$

Following DETR (Carion et al. 2020), we first perform bipartite matching between the predicted set $\hat{\mathbf{y}}$ and the ground

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| Configurations | AVA | | | UCF | JHMDB |
|---|---|---|---|---|---|
| | SF-R50 | SF-R101 | ViT-B | SF-R101 | SF-R101 |
| Optimizer | AdamW | | | | |
| Initial learning rate | 1e-5 (backbone), 1e-4 (all others) | | | | |
| Weight decay | 1e-4 | | | | |
| Optimizer momentum | $\beta_1$, $\beta_2$ = 0.9, 0.999 | | | | |
| Learning rate schedule | step decay (decay rate = 0.1) | | | | |
| Batch size | 8 | 16 | 16 | 32 | 32 |
| Decay step | 5 | 7 | 7 | 1 | 1 |
| Training epochs | 6 | 8 | 8 | 3 | 3 |
| Training input resolution | 256 | 256 | 224 | 224 | 256 |
| Inference input resolution | 256 | 256 | 224 | 256 | 256 |

Table 3: Training settings and input resolutions for JoVALE.

truth set $\mathbf{y^{gt}}$. The matching cost $\mathcal{L}_{\text{match}}$ is defined as

$$\begin{aligned}\mathcal{L}_{\text{match}}(y_i^{gt}, \hat{y}_j) =& \mathbb{1}_{\{h_i^{gt} \neq \varnothing\}} \mathcal{L}_{\text{cls}}(h_i^{gt}, h_j) \\ &+ \mathbb{1}_{\{h_i^{gt} \neq \varnothing\}} \mathcal{L}_{\text{box}}(b_i^{gt}, b_j),\end{aligned} \tag{2}$$

where $\mathcal{L}_{\text{cls}}$ is the sigmoid focal loss (Lin et al. 2017), and $\mathcal{L}_{\text{box}}$ is a combination of $l1$ loss and GIoU loss (Rezatofighi et al. 2019). The optimal matching $\hat{\sigma}$ is obtained by minimizing the total matching cost as

$$\hat{\sigma} = \underset{\sigma \in \mathfrak{S}_K}{\operatorname{argmin}} \sum_{i=1}^{K} \mathcal{L}_{\text{match}}(y_{\sigma(i)}^{gt}, \hat{y}_i). \tag{3}$$

Once the optimal matching is established, we calculate the action classification loss

$$\mathcal{L}(\mathbf{y^{gt}}, \hat{\mathbf{y}}) = \sum_{i=1}^{K} \mathcal{L}_{\text{cls}}(c_{\hat{\sigma}(i)}^{gt}, c_i). \tag{4}$$

# Additional Experiment

## Effect of Scene-descriptive Features

In Table 4, we evaluate the impact of different scene-descriptive features on the JoVALE model. The baseline model is a version of JoVALE that leverages only audio and visual features, without utilizing scene-descriptive features. Incorporating scene-descriptive features derived from vision-language pre-trained models, such as CLIP (Radford et al. 2021) and BLIP (Li et al. 2022), significantly boosts performance to 32.0 and 32.6 mAP, respectively. Further enhancement is observed with BLIP*, which refers to BLIP fine-tuned for image captioning, increasing the performance to 34.0 mAP.

| | Baseline | CLIP | BLIP | BLIP* |
|---|---|---|---|---|
| mAP | 28.6 | 32.0 | 32.6 | **34.0** |

Table 4: Effect of scene-descriptive features. BLIP marked with * denotes the BLIP image captioning model.

## Multi-modal Fusion Strategies

## Per-category Analysis

Fig. 1 shows the per-category average precision of the multi-modal and visual-only versions of JoVALE to analyze the performance gains from using multi-modal features. The blue bars represent the visual-only version of JoVALE, while the orange bars correspond to the multi-modal version. Both models, configured with SlowFast-R50, were evaluated on the AVA v2.2 dataset. The results demonstrate significant improvements in categories where audio cues are critical, such as 'sing,' 'play musical instrument,' and 'answer phone,' highlighting the effectiveness of integrating multi-modal information.

# Visualization

Fig. 2 presents activation maps comparing the multi-modal JoVALE, which leverages audio, visual, and scene-descriptive features, with the visual-only JoVALE on the AVA v2.2 dataset. The left three columns display heatmaps from the multi-modal JoVALE, while the right three columns show those from the visual-only JoVALE. Each pair of rows corresponds to video clips labeled 'work on a computer,' 'answer phone,' and 'read.'

The multi-modal JoVALE demonstrates an enhanced ability to focus on key objects relevant to the actions, such as the keyboard, phone, and note, particularly in the final layer by integrating audio, visual, and contextual scene information. This integration allows the model to capture more nuanced aspects of the scene that are critical for accurate action recognition. In contrast, the visual-only JoVALE predominantly focuses on human regions, which suggests a limitation in understanding the full context of the scene when relying solely on visual features. The qualitative results highlight the advantage of multi-modal fusion in capturing comprehensive scene information, leading to more precise and context-aware action detection.

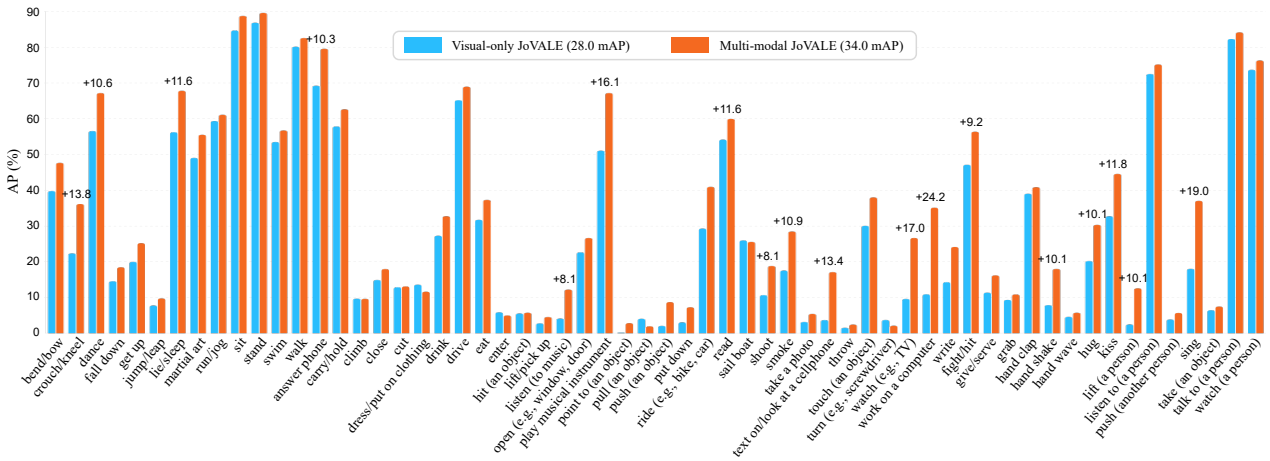


Figure 1: Comparison of per-category performance between the visual-only and multi-modal JoVALE.

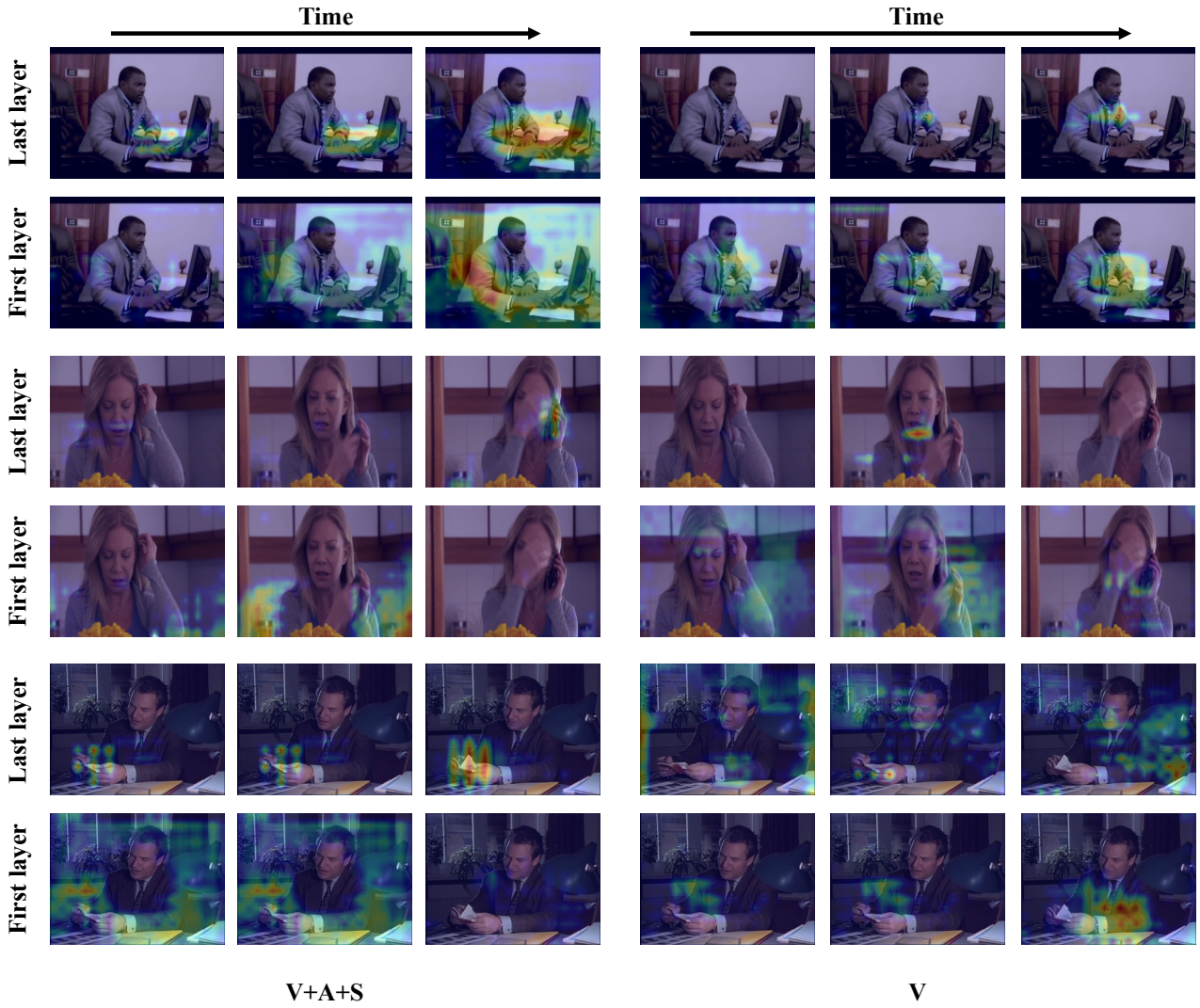


Figure 2: Additional visualization of activation maps. The left side shows heatmaps for the JoVALE model utilizing audio, visual, and scene-descriptive features, while the right side shows heatmaps for the JoVALE model using only visual features.

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